Food Delivery Demand Pulse — Executive Summary

Investigation period: Jan–Mar 2025 · 50,000 orders · 7 cities · 90 days

---

The core finding in one sentence

The current surge policy is paying rider incentives reactively — *after* queues form — and it is not reducing delivery times. Surge orders take **3.7 minutes longer** to deliver than non-surge orders (43.2 min vs 39.5 min, $r = 0.12$, $p < 0.001$).

What the data actually shows

Metric	Finding
True peak hours	12–13:00 (lunch) and 19–21:00 (dinner) — 44% of all orders in 5 hours
Off-peak surge waste	12.7% of surge orders fall outside peak hours (1,515 orders in 90 days)
Weekend surge rate	32.2% vs 20.5% weekday — but weekend delivery is 4.5 min slower
Hour 18 blind spot	3,683 orders/hour with only 5.7% surge — same as 3am
Surge effectiveness	No measurable delivery-time improvement at any peak hour
City cohorts	All 7 cities peak identically (hours 19, 20, 13). Uniform policy is correct by accident.

---

3 Recommendations

Recommendation 1 — Kill off-peak surge immediately

Action: Restrict surge to 12:00–13:59 and 19:00–21:59 only. Remove surge from all other hours.

Why: 505 surge orders/month fall outside true peak windows. These hours have idle rider supply — surge adds cost without any demand-supply benefit.

Estimated monthly saving: ₹15,000–₹25,000 in rider incentives **Risk level:** Low. No demand change, no delivery time impact. **Time to implement:** 1 day (config change).

Recommendation 2 — Add a weekend pre-position window at 18:00

Action: On Friday–Sunday, activate a ₹15 per-rider pre-positioning bonus at 18:00, ahead of the dinner surge.

Why: Weekends have 11.4pp higher surge rates and 4.5 min longer deliveries vs weekdays. Hour 18 receives 1,112 weekend orders but no surge signal. Riders arrive after queues form, not before.

Estimated delivery improvement: −2 to −4 min on weekend evenings **A/B test design:** 3 treatment cities (Kolkata, Delhi, Mumbai) vs 4 control cities, 2-week run on Fri–Sun only. ~1,400 orders = sufficient statistical power. Primary metric: p50 weekend delivery time. **Risk level:** Low-medium. Small incentive cost increase (est. ₹8,000–12,000/month) with measurable upside.

Recommendation 3 — Pilot pre-positioning model in Kolkata to replace reactive surge

Action: Replace ₹30–50 reactive surge bonus with a ₹20 pre-position bonus paid to riders who check in to designated zones by 12:00 and 18:45 daily. Pilot city: Kolkata.

Why this city: Kolkata has the highest demand concentration (32.1% of orders in just 3 hours) — most to gain from supply certainty. The data shows surge incentives attract riders *after* the queue peaks, not before. The baseline delivery time in low-volume hours is 39.5 min vs 43.2 min during surge — a 3.7-min penalty that pre-positioning can eliminate.

Estimated delivery improvement: −4 to −6 min at peak hours **Estimated cost:** Neutral to slightly cheaper than current reactive surge if pre-position rate < 60% of current surge rate **Pilot timeline:** 2 weeks to measure. Scale in 4 weeks if Δdelivery > 2 min at $p < 0.05$. **Risk level:** Medium. Requires dispatch team coordination. Low financial risk.

7-Day Forecast — Delhi

Date	Forecast	Lower	Upper
Apr 1 (Tue)	90 orders	76	103
Apr 2 (Wed)	90 orders	76	103
Apr 3 (Thu)	90 orders	76	103
Apr 4 (Fri)	90 orders	76	103
Apr 5 (Sat)	90 orders	76	103
Apr 6 (Sun)	90 orders	76	103
Apr 7 (Mon)	90 orders	76	103

Model: 14-day trailing average with ±15% uncertainty band. **Production evaluation metric:** MAPE (Mean Absolute % Error). Target < 15% for staffing decisions. If 3-day rolling MAPE exceeds 20%, refit model. Recommended upgrade path: ARIMA(1,1,1) or Prophet with a binary "rain" feature from IMD Open API (estimated −4–6pp MAPE reduction).

Combined Impact Summary

Recommendation	Monthly saving / benefit	Confidence
Rec 1: Narrow surge hours	₹15,000–₹25,000 cost saving	High
Rec 2: Weekend pre-position	−2 to −4 min delivery (weekends)	Medium (needs A/B)
Rec 3: Pre-position pilot (Kolkata)	−4 to −6 min delivery at peak	Medium (pilot required)
Combined	**₹15K–25K saving + ~5 min delivery improvement**	—

Analysis by: Data Science investigation team · Case 3 · Q1 2025 Dataset: case3_food_delivery_orders.csv (50,000 rows, synthetic, Jan–Mar 2025)